

CitySim: One Data Pipeline, Three Cities: Building MATSim Mobility Digital Twins for Pedestrian Safety, Flood Resilience, and Event Traffic Management

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1. Background and Motivation

CityData.AI builds mobility digital twins by combining large-scale passive mobility data with agent-based simulation. Our data platform, CityFlow, collects, curates, and analyses mobility big data from connected-vehicle traces, mobile-phone-derived movement, and related sources, to produce origin-destination matrices, full-day, multimodal trip datasets (driving, walking, cycling, and public transit), and patterns of life (visitation and stay patterns for places of interest). These datasets are then converted into MATSim plans and executed as full-scale, high-resolution simulations, giving us a repeatable pipeline: raw mobility data through CityFlow, into a MATSim digital twin, and out as decision-support analytics for public agencies and planners. We've named this pipeline "CitySim".

This abstract presents three applications of that pipeline across very different problem classes and geographies. Two are completed studies with quantitative results (Varanasi and Port of Spain); the third (Miami-Dade) is an ongoing deployment we present as a forward-looking use case. Together they illustrate how a single MATSim-centred workflow adapts to pedestrian microsimulation, disaster resilience, and large-event traffic management.

2. Varanasi, India — Predicting Pedestrian Overcrowding

Problem. Varanasi's historic core has narrow, high-footfall corridors and major religious destinations where crowd conditions can become dangerous. Direct measurement is sparse, and the operational question is not simply "what will one camera count be?" but "what network-consistent pedestrian state is most plausible given what has been observed so far?"

Approach. We developed a three-layer, simulation-driven forecasting system. A demand-construction process produces full-day mobility traces that are converted into MATSim plans. A MATSim physical replay layer routes those plans on a walk-focused pedestrian network, using a network walk mode, QSim with PassingQ link dynamics at one-second time steps, capacities scaled to sample size, and a width-aware routing disutility that penalises narrow lanes. A causal, nonnegative-blend estimator then infers the current day as a nonnegative combination of historical replay days from the camera observations available up to the issue time. Conceptually this is closer to data assimilation and analog forecasting than to black-box extrapolation: the goal is to recover a coherent latent network state so that link flows, densities, and multi-site forecasts remain physically and behaviourally consistent.

Results. The prototype was backtested on 4 active camera sites (of 7 now deployed), across 7 replay days and 140 causal, day-wise backtest contexts. Under strict causal backtesting:

- Tier 1 is the stronger model. The observation-space blend forecast beats the interpretable Tier 2 replay-to-camera ridge model by roughly 23% on mean absolute error (Tier 1 MAE 23.26, RMSE 34.78; Tier 2 MAE 30.18, RMSE 41.73).

- The error is small relative to the signal. Tier 1 forecast error is of the same order as the intrinsic short-term variability of the raw camera counts (median raw-prefix RMSE around 14.85) — the model tracks the observed pedestrian signal to within a small multiple of its own noise rather than diverging from it.
- Accuracy is strongest in the calmer early-morning hours, and degrades as daytime pedestrian volumes and volatility rise.
- The blend behaves sensibly. Diagnostics point to a moderate-mixture regime rather than collapse onto a single day (mean active support around 3.6 of 7 replay days; normalised entropy around 0.56 on a 0–1 scale).

Taken together, the latent blended-replay concept is empirically supported at small scale, while the observation model mapping replay outputs back to camera counts remains the clear priority for improvement.

3. Port of Spain, Trinidad & Tobago — Flood Resilience

Problem. We assessed the resilience of the Port of Spain transport network against a significant flood event, focusing on whether the disruption of a few critical links would propagate into system-wide failure.

Approach. Using CityFlow, we collected and curated an entire month of mobility big data for the region and built origin-destination matrices and a multimodal trips dataset. This fed a full-scale, minute-by-minute agent-based MATSim simulation of the population in and around Port of Spain—a Mobility Digital Twin of the city. We compared a typical weekday baseline against a flooded scenario in which critical, flood-prone road segments in the city centre were rendered unusable.

Results. The network proved highly vulnerable to cascading failure:

- Average travel time rose 41% (nearly 31 minutes added per trip) while average distance rose only 1.2% — a non-linear tipping point where rerouted traffic overwhelmed the remaining network.
- Completed trips fell 3.4%, meaning congestion made thousands of journeys effectively impossible.
- Macroscopic Fundamental Diagram analysis showed the network's peak morning arrival rate fell nearly 10%, and its ability to recover from congestion dropped about 35%.
- A pronounced hysteresis effect meant the system failed to recover between morning and evening peaks, so unresolved morning congestion compounded the evening peak.

Takeaway. Flooding pushes the network into gridlock easily but lets it recover only slowly. The impact is a systemic breakdown in mobility and access, not a simple inconvenience. Findings support hardening critical corridors, adding network redundancy, promoting multimodal transport, and developing dynamic, data-driven traffic management.

4. Miami-Dade County, Florida, USA — Major-Event Traffic Management (Ongoing)

Problem. Major events at Hard Rock Stadium—Copa América, FIFA World Cup matches, Formula 1, and similar—generate large, concentrated demand surges that strain the surrounding network. Local governments need a way to test interventions before committing to them.

Approach and goal. We are extending the same CityFlow-to-MATSim pipeline to Miami-Dade to build an event-focused digital twin. The objective is to let the county simulate and compare intervention scenarios—road closures, rerouting, and other traffic-management measures—under event conditions, so mitigation strategies can be evaluated *in silico* rather than learned during the event itself. This project is ongoing and we do not yet have quantitative results; we present it as a forward-looking application and welcome community input on event-demand modelling and intervention design.

5. Discussion and Contribution

Across three very different problems—pedestrian crowding, flood resilience, and event traffic—the same underlying workflow applies: passive mobility big data curated through CityFlow, converted into MATSim plans, and executed as a full-scale digital twin whose outputs drive decision support. The cases span scales (dense pedestrian core to metropolitan road network) and modes (network walking to multimodal vehicular flow), and they show MATSim serving both as a forecasting engine constrained by live observations (Varanasi) and as a scenario-comparison tool for resilience and intervention testing (Port of Spain, Miami-Dade). We hope to discuss with the community the practical challenges of grounding data-driven demand in MATSim, calibrating pedestrian dynamics, and validating scenario outputs for public-sector decision-making.